

Thesis progress report

LLM-Guided Neural Network Compilation for Real-time UAV Edge Inference

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A plan can be refused

The seven models can be compiled. This report is only the gate that sits in front of that rewrite.

A plan is JSON. It holds at most eight steps. Each step name must come from a closed list. The plan may also set how hard ONNX Runtime optimizes the graph, how many threads to use, and whether execution is sequential or parallel.

An unknown name rejects the whole plan. The engine never sees it. A known name that does not match this graph, such as batch-norm fusion when the graph has no batch norm, is dropped. The rest of the plan continues.

The checker allows three tries. If the reply is still illegal, an offline heuristic plan is used. That result is labelled as a fallback.

On this CPU, a thread count above the number of cores is cut down to that number. Parallel execution is left as the plan asked for it. The checker does not rename the plan. It only removes or clamps what cannot run.

Crashing on a useless step was the first checker. It made ordinary YOLO plans look broken, because a batch-norm fusion is legal in the list and absent in those graphs. A crash cannot be compared with another plan. A drop can. An unknown name is different: that is the model leaving the list, so the whole plan is refused. After three bad replies the offline plan is used, and it has to be labelled as a fallback. An earlier table called an 8-bit heuristic plan by a fusion name. The name lied, and the table could not be read. The name now has to match the steps.

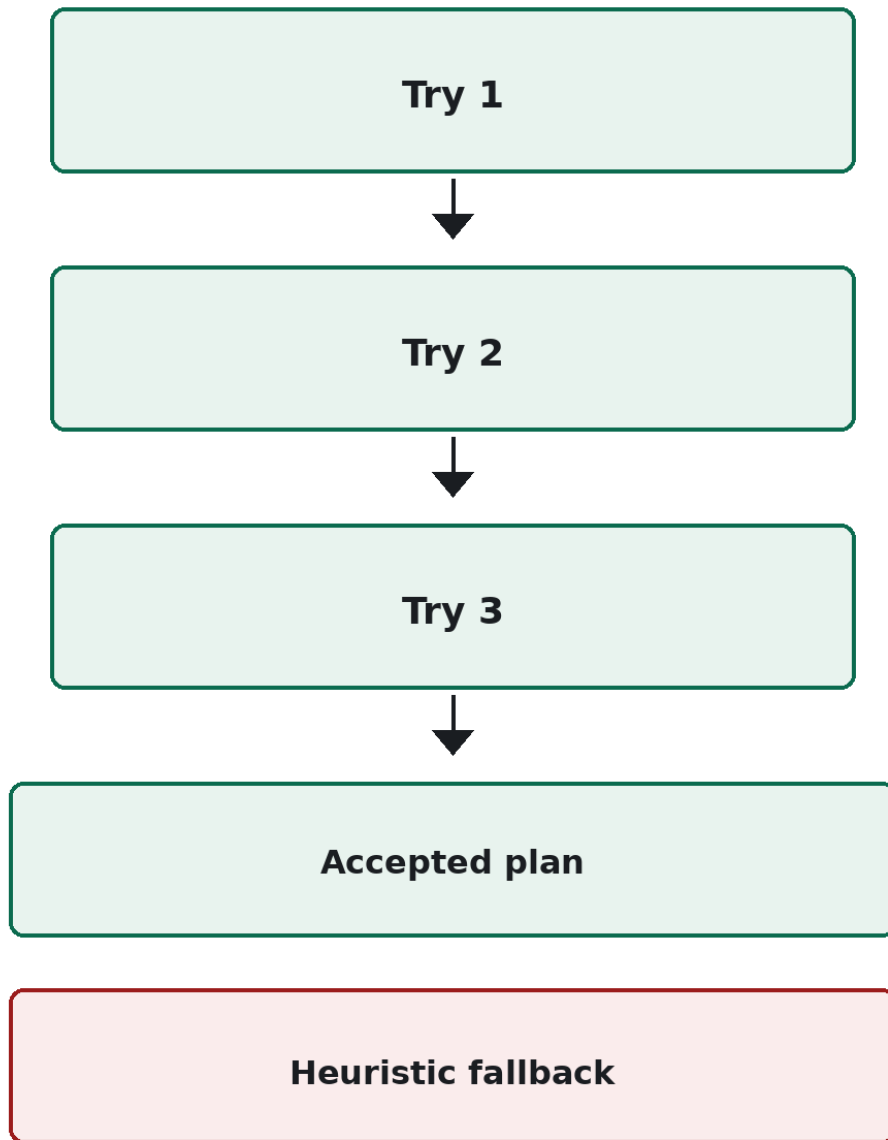


Figure 1. Three tries. Then an accepted plan, or the heuristic fallback.

An illegal plan is rejected or cut before the engine runs. Which of the legal plans is kept is decided only after they are measured.